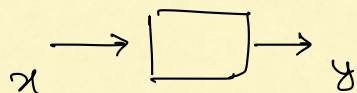


Q.1



$$y(t) = x(t - t_0)$$

$$x_1 \rightarrow \boxed{} \rightarrow y_1 = x_1(t - t_0)$$

$$x_2 \rightarrow \boxed{} \rightarrow y_2 = x_2(t - t_0).$$

$$x_1 + x_2 \rightarrow \boxed{} \rightarrow z. \quad \text{check if } z = y_1 + y_2.$$

In this case the output is the time shifted version of $x_1 + x_2$, i.e., the value of $x_1 + x_2$ at time $t - t_0$

$$= x_1(t - t_0) + x_2(t - t_0)$$

$$= y_1(t) + y_2(t)$$

$\therefore$ Additivity holds.

Now, suppose

$$x \rightarrow \boxed{} \rightarrow y, \quad y(t) = x(t - t_0)$$

then,

$$\alpha x \rightarrow \boxed{} \rightarrow z. \quad \text{check if } z = \alpha y$$

z = The output is time shifted version of $\alpha x(t)$

$$= \alpha \cdot x(t - t_0)$$

$$= \alpha y(t)$$

$\therefore$ Homogeneity holds.

$\Rightarrow$ This system is linear.

Q2. $x \rightarrow \boxed{} \rightarrow y$ with $y(t) = x(t) + a$, $a \neq 0$.

$$ax \rightarrow \boxed{} \rightarrow z$$

Check if $z = ay$:

$$z(t) = \text{input} + a = ax(t) + a$$

$$\text{But } ay(t) = ax(t) + aa$$

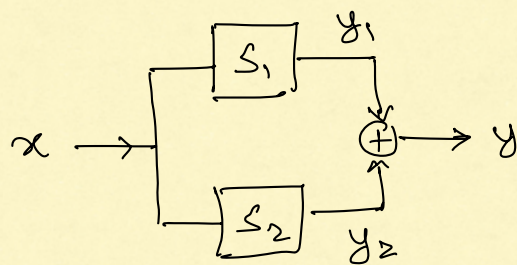
$$\text{Clearly } z(t) \neq ay(t).$$

Homogeneity fails.

$\Rightarrow$ System is not linear.

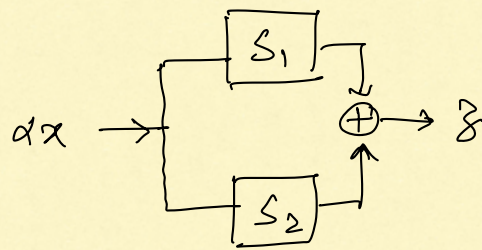
Remark: In fact, additivity also fails for this system.

Q4: Parallel Interconnection



Both S_1, S_2 are linear. So both satisfy additivity and homogeneity.

Homogeneity for overall system:



check if $z = \alpha y$.

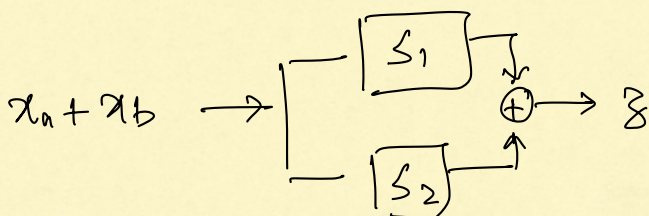
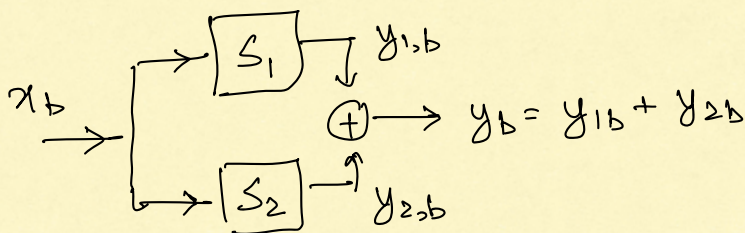
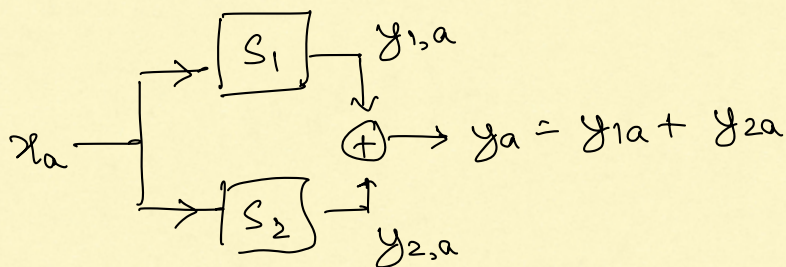
Output of $S_1 = \alpha y_1$ (because S_1 is linear)

Output of $S_2 = \alpha y_2$ (" " " ")

$$\therefore z = \alpha y_1 + \alpha y_2 = \alpha (y_1 + y_2) \\ = \alpha y.$$

$\therefore$ Homogeneity is satisfied.

Additivity for overall system:



check if $z = y_a + y_b$

Since S_1 is linear:

$$\text{its output is } = y_{1a} + y_{1b}$$

Since S_2 is linear:

$$\text{its output is } = y_{2a} + y_{2b}$$

$$\begin{aligned}\therefore Z &= y_{1a} + y_{1b} + y_{2a} + y_{2b} \\ &= (y_{1a} + y_{2a}) + (y_{1b} + y_{2b}) \\ &= y_a + y_b.\end{aligned}$$

$\therefore$ Additivity holds for this system.

$\therefore$ This system is linear.

Q.6



$$\sum_{k=0}^n a_k \cdot y^{(k)}(t) = x(t).$$

Homogeneity:

$$\alpha x \rightarrow \boxed{\text{ }} \rightarrow Z \quad \text{check if } Z = \alpha y.$$

We will simply check if $Z = \alpha y$ solves

$$\sum_{k=0}^n a_k Z^{(k)}(t) = \alpha \cdot x(t).$$

$$\text{Note that } Z^{(k)}(t) = \alpha \cdot y^{(k)}(t)$$

$$\therefore \sum_k a_k Z^{(k)}(t) = \alpha \sum_k a_k \cdot y^{(k)}(t) = \alpha x(t).$$

$\therefore$ Homogeneity holds.

Additivity:

$$x_1 \longrightarrow \boxed{} \longrightarrow y_1 \quad \sum a_k \cdot y_1^{(k)}(t) = x_1(t)$$

$$x_2 \longrightarrow \boxed{} \longrightarrow y_2 \quad \sum a_k y_2^{(k)}(t) = x_2(t)$$

$$x_1 + x_2 \longrightarrow \boxed{} \longrightarrow z, \quad \text{Check if } z = y_1 + y_2.$$

Again, we will simply check if $z = y_1 + y_2$ solves

$$\sum_k a_k z^{(k)}(t) = x_1 + x_2.$$

$$\text{Note that } z^{(k)}(t) = y_1^{(k)}(t) + y_2^{(k)}(t)$$

$$\begin{aligned} \therefore \sum_k a_k z^{(k)}(t) &= \sum_k a_k y_1^{(k)}(t) + \sum_k a_k y_2^{(k)}(t) \\ &= x_1(t) + x_2(t). \end{aligned}$$

$\therefore$ Additivity holds for this system.

$\therefore$ This system is linear.

Q. 7(c) : HINT:

Consider a time shift operator system
or a system that integrates its input.